ASR-11 Primary Surveillance Radar

Representative parameter datasheet for MATLAB agentic-workflow exercises

Revision: 1.0 · 25 August 2026
Classification: Training Use Only

IMPORTANT — SIMULATION DATA ONLY. This is a fictional, representative training artifact inspired by the public description of an ASR-11-class terminal surveillance radar. Values, site name, and configuration are invented for classroom modeling; they are not manufacturer, FAA, airport, or operational specifications and must not be used for navigation, procurement, safety, or real-world system configuration.

Terminal-area primary surveillance

Rotating L-band radar configuration designed as a coherent parameter set for a classroom air-traffic-control scenario and a first-pass MATLAB radar simulation.

60	12.5	2.8
NMI INSTRUMENTED RANGE	RPM SCAN RATE	GHZ NOMINAL FREQUENCY

Configuration at a glance

Radar role	**Primary surveillance**
Scenario ID	**KMTN-TRN-01**
Coverage mode	**Terminal / 360°**
Propagation model	**Free-space baseline**
Target class	**Fixed-wing aircraft**
Configuration	**Baseline terminal coverage**

1. System identity and operating envelope

Parameter	Representative value
System designation	ASR-11 (training representative)
Radar type	Coherent pulsed primary surveillance radar
Frequency band	L-band (training convention)
Instrumented range	1.5–60 nmi
Azimuth coverage	0–360°
Elevation coverage	0–30°
Environmental baseline	Standard atmosphere; clear air

2. Primary model input set

Parameter	Representative value
Carrier frequency, fc	2.80 GHz
Peak transmit power, Pt	1.10 MW
Pulse repetition frequency, PRF	Selectable / waveform-dependent; 1,000 Hz baseline
Transmit pulse width, τ	0.80 µs
Antenna peak gain, G	34 dBi
Azimuth 3 dB beamwidth	1.4°
Elevation 3 dB beamwidth	4.5°
Antenna scan rate	12.5 rpm
Receiver noise figure	4.0 dB
System losses	7.0 dB

Reference target RCS	**10 m²**

ASR-11 Modeling Notes

3. Detection and measurement assumptions

Parameter	Representative value	Use in exercise
Probability of detection, Pd	0.90 at reference condition	Reference only: 10 m^2 target at 50 nmi, boresight, clear air.
Probability of false alarm, Pfa	1.0e-6 per resolution cell	Set detector threshold or use as a stated evaluation constraint.
Range measurement accuracy	±90 m (1σ)	Add zero-mean Gaussian range error for track reports.
Azimuth measurement accuracy	±0.12° (1σ)	Add zero-mean Gaussian bearing error for track reports.
Range resolution (nominal)	120 m	Use cτ/2; no pulse compression in baseline configuration.
Unambiguous range (baseline)	150 km	At the 1,000 Hz baseline waveform; exceeds the 60 nmi reporting gate.
Update interval	4.8 s	One measurement opportunity per antenna revolution.

4. Classroom air-traffic-control scenario

Radar site

Fictional KMTN Training Terminal, reference position [0, 0, 0] m in local ENU coordinates. Antenna phase center height: 15 m above ground level.

Targets at scenario start

Track	Position [E, N, U] m	Velocity [E, N, U] m/s	RCS
TRN101	[18,000, 65,000, 2,800]	[−115, −40, 0]	10 m^2
TRN205	[−32,000, 48,000, 4,100]	[95, −65, −2]	6 m^2
TRN317	[7,000, 26,000, 1,700]	[45, 130, 3]	3 m^2

Exercise scope

- Import this document and extract a structured parameter set.
- Build a rotating monostatic radar model.
- Simulate range/azimuth detections for the three targets.
- Compare ideal versus noisy measurement reports over 60 s.

Deliberately excluded from baseline

secondary surveillance · IFF · terrain masking · weather clutter · multipath · target fluctuation

5. Data-quality notes

Rule	Expected behavior
Operating units	Values are stated in the customary units used in this reference document.
Derived quantities	Range resolution, unambiguous range, and scan period are nominal values based on the listed baseline configuration.
Configuration control	Changes to the listed values require a corresponding revision to this training reference.
Document status	Fictional training reference; not an operational, procurement, or maintenance specification.

ASR-11 is referenced only to provide a recognizable classroom context. This document does not represent a manufacturer, FAA, airport, or operational specification.